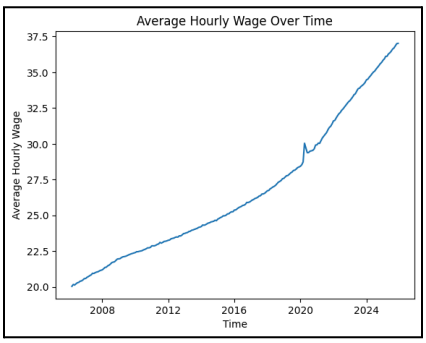
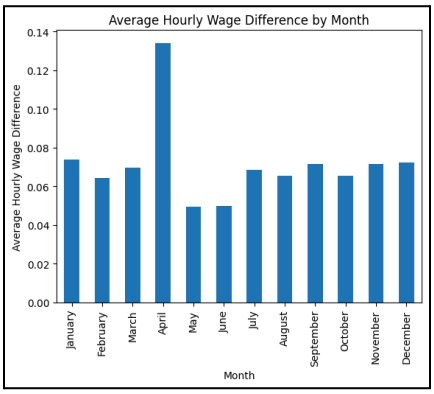
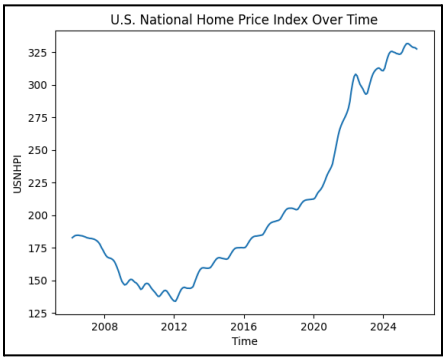
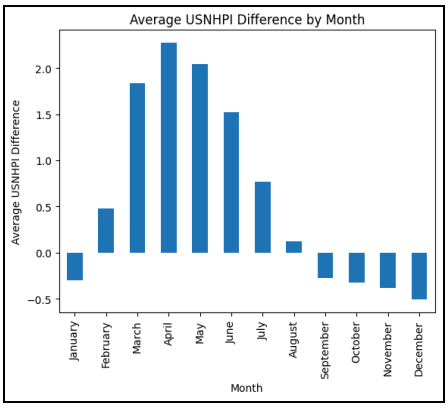
USNHPI and Average Hourly Wage Dataset:

For the dataset we are using for project 2, the unit of observation is each month dating back to March of 2006 (20 years ago) that is associated with a corresponding U.S. National Home Price Index value as well as the average hourly wage in the United States during that month.

- *date:* The *date* variable is the month corresponding to each USNHPI value and average hourly wage value.
    - It is important to point out that our data set contains 238 total observations (or months) ranging from March 2006 through December 2025.
- *avg_hourly_wage:* The *avg_hourly_wage* variable highlights the average hourly wage in the United States in a given month.
    - 

- *avg_hourly_wage_lag:* The *avg_hourly_wage_lag* variable is a mutation of the *avg_hourly_wage* variable that lags average hourly wage by one month since we are trying to use the previous month's average hourly wage as the measure to predict the activity of the USNHPI.
    - Example: The average hourly wage value for March 2006 will be linked to the USNHPI value from April 2006.
- *avg_hourly_wage_diff:* The *avg_hourly_wage_diff* variable is a mutation of the *avg_hourly_wage* variable that looks into the change in average hourly wage from the previous month in order to identify possible trends based on time of the year.
- *avg_hourly_wage_log_diff:* The *avg_hourly_wage_log_diff* variable is a mutation of the *avg_hourly_wage* variable that log-transforms the data to ensure it is stationary.
    - 

- *USNHPI:* The *USNHPI* variable represents the values of the U.S. National Home Price Index by month, which tracks the monthly change in the value of single-family homes across the entire United States.
  - 

- *USNHPI_diff:* The *USNHPI_diff* variable is a mutation of the *USNHPI* variable that looks into the change in the USNHPI from the previous month in order to identify possible trends based on time of the year.
  - 

- *USNHPI_log_diff:* The *USNHPI_log_diff* variable is a mutation of the *USNHPI* variable that log-transforms the data to ensure it is stationary.

Column	Description	Potential Responses
date	Date of observation	3/1/2006 - 12/1/2025
date_diff	Number of days over which wage and USNHPI were measured	28 days, 30 days, 31 days
avg_hourly_wage	Average hourly wage in the U.S. in given month	20.04, 33.02, 37.02
avg_hourly_wage_lag	Lagged average hourly wage to be used as predictor	20.04, 33.02, 37.02
avg_hourly_wage_diff	Shows the change in average hourly wage from previous month	0, -0.04, 0.11
avg_hourly_wage_log_diff	The log transform of average hourly	0.004460973940624946,

	wage difference	-0.001985112314420334
USNHPI	Raw index value for each month	182.748, 141.037, 327.455
USNHPI_diff	Shows change in U.S. National Home Price Index from previous month	0.9, -1.112, 4.998
USNHPI_log_diff	The log transform of USNHPI difference	0.003977963444496524, -0.0011002383152698414